Data Preprocessing & Feature Engineering

Overview

This notebook focuses on the feature engineering and data preparation process for the Bank Marketing dataset. After the initial exploration in the **EDA** phase, we now aim to transform raw data into a model-ready format. This involves creating new features, handling missing data, and performing necessary transformations to ensure the dataset is optimal for machine learning.

The goal is to refine the features in a way that enhances model performance while avoiding issues like data leakage or unnecessary redundancy.

Objectives

- Create core engineered features such as target variable encoding, binary flags, and ordinal variables
- Add derived behavioral features that reflect client engagement, like multiple contacts within a campaign
- Generate interaction features to capture complex relationships between variables (e.g., "many contacts and housing loan")
- Bin continuous variables into interpretable categories for improved model performance
- Review and select features based on their relevance to the target variable

Dataset Description

We start with a cleaned dataset that has already undergone an initial exploration. The raw dataset consists of **45,211 observations** and **17 features**, which include a mix of numerical and categorical data.

After preprocessing, the dataset will include engineered features that are tailored to help our machine learning models, including:

- **Binary features**: E.g., whether a client has a loan, housing, or was contacted in a previous campaign
- **Categorical features**: Ordinal variables like education level, and newly binned categories (e.g., age group, balance bucket)
- **Interaction features**: Multiplicative combinations of existing features, such as combining multiple contacts with loan status

Key Considerations

- Handle **missing or implicit data**, such as the "unknown" values in categorical features and the sentinel value (-1) in `pdays`
- Avoid **data leakage** (e.g., excluding `duration` from the feature set, which is only known after the call)
- Create **interpretable features** by binning continuous variables and generating categorical labels that are useful for downstream modeling

Outcome

By the end of this notebook, the dataset will be enriched with engineered features, ready to be passed into the modeling pipeline. These features will be optimized for machine learning algorithms, helping us build more accurate and insightful models in the next step. The train, validation and test sets that will be used in the modelling notebook will be created.

```
# quarto preview 02_preprocessing.ipynb --to pdf
# quarto render 02_preprocessing.ipynb
# black 02_preprocessing.ipynb
```

```
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.model_selection import train_test_split

sns.set_theme(style="whitegrid", palette="deep")
pd.set_option("display.max_columns", 25)
```

1. Feature engineering

1. Create core engineered features

Create the main cleaned and engineered feature set.

```
nb = "01"
data_dir = Path("data")

df_path = data_dir / f"{nb}_raw_data.parquet"

df_ = pd.read_parquet(df_path)
print(df_.shape)
df_.head(3)
```

```
(45211, 17)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	no	2143	yes	no	unknown	5	may	2
1	44	technician	single	secondary	no	29	yes	no	unknown	5	may	1
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown	5	may	7

```
processed = df_.copy()

processed["target"] = (processed["y"] == "yes").astype(np.int8)

processed["pdays_was_contacted"] = (processed["pdays"] != -1).astype(np.int8)
processed["pdays_since_prev"] = processed["pdays"].where(
    processed["pdays"] != -1, np.nan
)

month_order = [
    "jan",
    "feb",
    "mar",
    "apr",
    "may",
    "jun",
    "jul",
    "aug",
```

```
    "sep",
    "oct",
    "nov",
    "dec",
]
processed["month"] = pd.Categorical(
    processed["month"], categories=month_order, ordered=False
)

for c in ["default", "housing", "loan"]:
    processed[f"{c}"] = (processed[c] == "yes").astype(np.int8)
    # processed[f"{c}_binary"] = (processed[c] == "yes").astype(np.int8)

display(
    processed[
        [
            "target",
            "pdays_was_contacted",
            "pdays_since_prev",
            "month",
            "default",
            "housing",
            "loan",
        ]
    ].head()
)
```

	target	pdays_was_contacted	pdays_since_prev	month	default	housing	loan
0	0	0	NaN	may	0	1	0
1	0	0	NaN	may	0	1	0
2	0	0	NaN	may	0	1	1
3	0	0	NaN	may	0	1	0
4	0	0	NaN	may	0	0	0

Interpretation: Core engineered features convert raw columns into model-ready signals: `target` (int8 0/1 target), `pdays_was_contacted` (binary sentinel decoder), `month` as an unordered `pd.Categorical` for downstream OHE, and binary flags for `default`, `housing`, and `loan`.

```
print(processed.shape)
processed.head(3)
```

```
(45211, 20)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

2. Add non-overlapping derived behavioral features

Add a small set of additional behavioral features with low overlap with the core engineered features.

```
new_columns = []

# processed = processed.drop(
#     columns=["poutcome_failure", "history_info_missing"],
#     errors="ignore",
# )

processed["multiple_contacts_current_campaign"] = (processed["campaign"] > 1).astype(
    np.int8
)
new_columns.append("multiple_contacts_current_campaign")

processed["poutcome_success"] = (processed["poutcome"] == "success").astype(np.int8)
new_columns.append("poutcome_success")

processed["contact_is_unknown"] = (processed["contact"] == "unknown").astype(np.int8)
new_columns.append("contact_is_unknown")

processed["poutcome_is_unknown"] = (processed["poutcome"] == "unknown").astype(np.int8)
new_columns.append("poutcome_is_unknown")

print("Newly added columns:", new_columns)
display(processed[new_columns].head())
```

```
Newly added columns: ['multiple_contacts_current_campaign', 'poutcome_success', 'contact_is_u
```

	multiple_contacts_current_campaign	poutcome_success	contact_is_unknown	poutcome_is_unknow
0	0	0	1	1
1	0	0	1	1
2	0	0	1	1
3	0	0	1	1
4	0	0	1	1

Interpretation: Four non-overlapping behavioral indicators are added: `multiple_contacts_current_campaign` (whether the client was called more than once this campaign), `poutcome_success` (prior win), `contact_is_unknown` (no phone type on record), and `poutcome_is_unknown` (no prior campaign history). These distinguish engagement patterns without introducing redundancy with existing columns.

```
print(processed.shape)
processed.head(3)
```

```
(45211, 24)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	c
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

3. Add a small set of interaction features

Create compact interaction indicators from existing contact-history and loan-related feature columns.

```
new_columns = []

# processed = processed.drop(
#     columns=[
#         "contacted_before_and_success",
#         "contacted_before_and_failure",
#         "history_info_missing",
```

```
#     ],
#     errors="ignore",
# )

processed["many_contacts_and_housing_loan"] = (
    processed["multiple_contacts_current_campaign"] * processed["housing"]
).astype(np.int8)
new_columns.append("many_contacts_and_housing_loan")

processed["previous_and_many_current_contacts"] = (
    processed["pdays_was_contacted"] * processed["multiple_contacts_current_campaign"]
).astype(np.int8)
new_columns.append("previous_and_many_current_contacts")

processed["both_housing_and_personal_loan"] = (
    processed["housing"] * processed["loan"]
).astype(np.int8)
new_columns.append("both_housing_and_personal_loan")

print("Newly added columns:", new_columns)
display(processed[new_columns].head())
```

```
Newly added columns: ['many_contacts_and_housing_loan', 'previous_and_many_current_contacts'
```

	many_contacts_and_housing_loan	previous_and_many_current_contacts	both_housing_and_perso
0	0	0	0
1	0	0	0
2	0	0	1
3	0	0	0
4	0	0	0

Interpretation: Three interaction terms capture non-additive combinations: `many_contacts_and_housing_loa` (heavy campaign effort on a mortgaged client), `previous_and_many_current_contacts` (clients with both prior- and current-campaign contact history), and `both_housing_and_personal_loan` (heaviest debt burden). These multiplicative indicators add compound signal with minimal feature engineering overhead.

```
print(processed.shape)
processed.head(3)
```

```
(45211, 27)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

4. Add binned versions of selected numeric features

Bin `age`, `balance`, `campaign`, and `pdays_since_prev` into interpretable ordinal categories for downstream modelling.

```
new_columns = []

# age → age_group
processed["age_group"] = pd.cut(
    processed["age"],
    bins=[17, 25, 35, 50, np.inf],
    labels=["18-25", "26-35", "36-50", "51+"],
).astype(str)
new_columns.append("age_group")

# balance → balance_bucket
processed["balance_bucket"] = pd.cut(
    processed["balance"],
    bins=[-np.inf, 0, 500, 2000, np.inf],
    labels=["negative", "low", "medium", "high"],
    right=False,
).astype(str)
new_columns.append("balance_bucket")

# campaign → campaign_bucket
processed["campaign_bucket"] = pd.cut(
    processed["campaign"],
    bins=[0, 1, 3, np.inf],
    labels=["1", "2-3", "4+"],
).astype(str)
new_columns.append("campaign_bucket")

# pdays_since_prev → pdays_recency_bucket
```

```
processed["pdays_recency_bucket"] = np.select(
    [
        processed["pdays_was_contacted"] == 0,
        processed["pdays_since_prev"] <= 30,
        processed["pdays_since_prev"] <= 180,
    ],
    ["no_previous_contact", "recent", "medium"],
    default="old",
)
new_columns.append("pdays_recency_bucket")

print("Newly added columns:", new_columns)
display(processed[new_columns].head())
```

```
Newly added columns: ['age_group', 'balance_bucket', 'campaign_bucket', 'pdays_recency_bucke
```

	age_group	balance_bucket	campaign_bucket	pdays_recency_bucket
0	51+	high	1	no_previous_contact
1	36-50	low	1	no_previous_contact
2	26-35	low	1	no_previous_contact
3	36-50	medium	1	no_previous_contact
4	26-35	low	1	no_previous_contact

Interpretation: Binning converts right-skewed continuous variables into interpretable ordinal categories. `age_group` captures life-stage effects, `balance_bucket` handles negative values and the heavy positive tail, `campaign_bucket` reduces sensitivity to extreme contact counts, and `pdays_recency_bucket` converts the `-1` sentinel plus numeric days into a four-level recency label.

```
print(processed.shape)
processed.head(3)
```

```
(45211, 31)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	c
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

5. Review newly engineered features against the target

For each engineered feature that exists in `processed`, show a compact summary of category counts, proportions, and subscription rate vs `target`.

```
CANDIDATE_COLS = [
    "contact_is_unknown",
    "poutcome_is_unknown",
    "multiple_contacts_current_campaign",
    "poutcome_success",
    "many_contacts_and_housing_loan",
    "previous_and_many_current_contacts",
    "both_housing_and_personal_loan",
    "age_group",
    "balance_bucket",
    "campaign_bucket",
    "pdays_recency_bucket",
]
```

```
existing_candidate_cols = [c for c in CANDIDATE_COLS if c in processed.columns]

binary_check = pd.DataFrame(
    {"n_unique": [processed[c].nunique(dropna=False) for c in existing_candidate_cols]},
    index=existing_candidate_cols,
)

display(binary_check.sort_values("n_unique"))
```

	n_unique
contact_is_unknown	2
poutcome_is_unknown	2
multiple_contacts_current_campaign	2
poutcome_success	2
many_contacts_and_housing_loan	2
previous_and_many_current_contacts	2
both_housing_and_personal_loan	2

	n_unique
campaign_bucket	3
age_group	4
balance_bucket	4
pdays_recency_bucket	4

```
binary_cols = [
    "contact_is_unknown",
    "poutcome_is_unknown",
    "multiple_contacts_current_campaign",
    "poutcome_success",
    "many_contacts_and_housing_loan",
    "previous_and_many_current_contacts",
    "both_housing_and_personal_loan",
]

cat_cols = [
    "campaign_bucket",
    "age_group",
    "balance_bucket",
    "pdays_recency_bucket",
]
```

```
# Review binary engineered features by comparing counts and subscription rates across 0/1 gro
rows = []

for col in binary_cols:
    grp = processed.groupby(col)["target"].agg(count="count", subscription_rate="mean")
    grp.index = grp.index.map(lambda v: f"{col}={v}")
    rows.append(grp)

display(pd.concat(rows).round(3))
```

	count	subscription_rate
contact_is_unknown=0	32191	0.148
contact_is_unknown=1	13020	0.041
poutcome_is_unknown=0	8252	0.231
poutcome_is_unknown=1	36959	0.092
multiple_contacts_current_campaign=0	17544	0.146
multiple_contacts_current_campaign=1	27667	0.099

	count	subscription_rate
poutcome_success=0	43700	0.099
poutcome_success=1	1511	0.647
many_contacts_and_housing_loan=0	30199	0.141
many_contacts_and_housing_loan=1	15012	0.068
previous_and_many_current_contacts=0	40969	0.108
previous_and_many_current_contacts=1	4242	0.204
both_housing_and_personal_loan=0	40844	0.123
both_housing_and_personal_loan=1	4367	0.061

Interpretation: Features with a large subscription-rate gap between their 0 and 1 groups (e.g., `poutcome_success`) are the most discriminating individually. Features with smaller rate gaps (e.g., `both_housing_and_personal_loan`) may still be useful in combination with other predictors or for tree-based models that exploit interaction effects.

```
print(processed.shape)
processed.head(3)
```

```
(45211, 31)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

6. Define the final engineered feature set

Create a compact keep/drop split for engineered features based on the earlier review step.

```
candidate_keep = [
    "poutcome_success",
    "contact_is_unknown",
    "poutcome_is_unknown",
    "multiple_contacts_current_campaign",
    "previous_and_many_current_contacts",
    "many_contacts_and_housing_loan",
    "both_housing_and_personal_loan",
    "age_group",
```

```
    "balance_bucket",
    "campaign_bucket",
    "pdays_recency_bucket",
]

candidate_drop = ["y"]

keep_features = [c for c in candidate_keep if c in processed.columns]
drop_features = [c for c in candidate_drop if c in processed.columns]

print("Keep engineered features:")
print(keep_features)

print("\nDrop engineered features:")
print(drop_features)

print(f"\nNumber of kept engineered features: {len(keep_features)}")
print(f"Number of dropped engineered features: {len(drop_features)}")
```

```
Keep engineered features:
['poutcome_success', 'contact_is_unknown', 'poutcome_is_unknown', 'multiple_contacts_current_

Drop engineered features:
['y']

Number of kept engineered features: 11
Number of dropped engineered features: 1
```

Interpretation: All 11 engineered candidate features are kept — none were found sufficiently redundant or low-signal to remove at this stage. The `candidate_drop` list only contains `y` since we transformed it into a binary column (`target`), so the final modeling table will include all retained engineered features alongside the base columns.

```
print(processed.shape)
processed.head(3)
```

```
(45211, 31)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

7. Create the final modeling table

Save the dataset that contains the engineered features. We'll save the file containing all of the features, regardless of whether some might refer to the same information (ex. age and age bucket).

In the next section we'll save the datasets containing only the features that we'll actually use for modeling.

```
processed_final = processed.drop(
    columns=drop_features,
    errors="ignore",
).copy()

target_col = "target"
feature_cols = [c for c in processed_final.columns if c not in [target_col]]
```

```
print("Feature engineered data shape:", processed_final.shape)
processed_final.head(3)
```

```
Feature engineered data shape: (45211, 30)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

```
nb = "02"
data_dir = Path("data")

df_path = data_dir / f"{nb}_full_eng_data.parquet"

processed_final.to_parquet(df_path, index=False)
print(f"Saved engineered data: {df_path}")
```

```
Saved engineered data: data/02_full_eng_data.parquet
```

Interpretation: The final modeling table retains the full cleaned feature space before feature selection in Section 2.8.

8. Final feature list for pre-contact modeling

In this section, we'll select the features that we'll use for modeling.

```
nb = "02"
data_dir = Path("data")

df_path = data_dir / f"{nb}_full_eng_data.parquet"

df_ = pd.read_parquet(df_path)
print(df_.shape)
df_.head(3)
```

```
(45211, 30)
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	d
0	58	management	married	tertiary	0	2143	1	0	unknown	5	may	2
1	44	technician	single	secondary	0	29	1	0	unknown	5	may	1
2	33	entrepreneur	married	secondary	0	2	1	1	unknown	5	may	7

```
df_.columns
```

```
Index(['age', 'job', 'marital', 'education', 'default', 'balance', 'housing',
       'loan', 'contact', 'day', 'month', 'duration', 'campaign', 'pdays',
       'previous', 'poutcome', 'target', 'pdays_was_contacted',
       'pdays_since_prev', 'multiple_contacts_current_campaign',
       'poutcome_success', 'contact_is_unknown', 'poutcome_is_unknown',
       'many_contacts_and_housing_loan', 'previous_and_many_current_contacts',
       'both_housing_and_personal_loan', 'age_group', 'balance_bucket',
       'campaign_bucket', 'pdays_recency_bucket'],
      dtype='object')
```

```python
target = ["target"]

selected_features = [
    "job",
    "marital",
    "education",
    "day",
    "previous",
    "age_group",
    "balance_bucket",
    "campaign_bucket",
    "pdays_was_contacted",
    "pdays_recency_bucket",
    "month",
    "default",
    "housing",
    "loan",
    "poutcome_success",
    "contact_is_unknown",
    "poutcome_is_unknown",
    "multiple_contacts_current_campaign",
    "previous_and_many_current_contacts",
    "many_contacts_and_housing_loan",
    "both_housing_and_personal_loan",
]

excluded_cols = df_.columns.difference(selected_features + ["target"]).tolist()

print(f"Selected features: {selected_features}")
print(f"\nNumber of selected features: {len(selected_features)}")

print(f"\nExcluded columns: {excluded_cols}")
print(f"\nNumber of excluded features: {len(excluded_cols)}")
```

```
Selected features: ['job', 'marital', 'education', 'day', 'previous', 'age_group', 'balance_b

Number of selected features: 21

Excluded columns: ['age', 'balance', 'campaign', 'contact', 'duration', 'pdays', 'pdays_since

Number of excluded features: 8
```

```
df = df_[selected_features + target]
print("Selected features data shape:", df.shape)
df.head(3)
```

```
Selected features data shape: (45211, 22)
```

	job	marital	education	day	previous	age_group	balance_bucket	campaign_bucket	p
0	management	married	tertiary	5	0	51+	high	1	0
1	technician	single	secondary	5	0	36-50	low	1	0
2	entrepreneur	married	secondary	5	0	26-35	low	1	0

```
nb = "02"
data_dir = Path("data")

df_path = data_dir / f"{nb}_raw_data.parquet"
df.to_parquet(df_path, index=False)
print(f"Saved engineered data: {df_path}")
```

```
Saved engineered data: data/02_raw_data.parquet
```

Interpretation: 21 features are selected, spanning demographics (`job`, `marital`, `age_group`), financial state (`balance_bucket`, `default_binary`, `housing_binary`, `loan_binary`), campaign history (`day`, `previous`, `month`, `campaign_bucket`, `pdays_was_contacted`, `pdays_recency_bucket`), and behavioral/interaction signals. Intermediate derivations such as `pdays_since_prev` and raw `age` are excluded — they are either subsumed by bucketed versions or dropped as redundant.

2. Data Preprocessing

```
RANDOM_STATE = 705
test_size = 0.2
target = "target"
```

```
nb = "02"
data_dir = Path("data")

df_path = data_dir / f"{nb}_raw_data.parquet"
```

```
df_ = pd.read_parquet(df_path)
print(df_.shape)
df_.head(3)
```

```
(45211, 22)
```

	job	marital	education	day	previous	age_group	balance_bucket	campaign_bucket	p
0	management	married	tertiary	5	0	51+	high	1	0
1	technician	single	secondary	5	0	36-50	low	1	0
2	entrepreneur	married	secondary	5	0	26-35	low	1	0

1. Define preprocessing column groups

```
categorical_cols = [
    "job",
    "marital",
    "education",
    "age_group",
    "balance_bucket",
    "campaign_bucket",
    "pdays_recency_bucket",
    "month",
]

binary_cols = [
    "pdays_was_contacted",
    "default",
    "housing",
    "loan",
    "poutcome_success",
    "contact_is_unknown",
    "poutcome_is_unknown",
    "multiple_contacts_current_campaign",
    "previous_and_many_current_contacts",
    "many_contacts_and_housing_loan",
    "both_housing_and_personal_loan",
]
```

```
numerical_cols = [
    "day",
    "previous",
]

print(f"{len(categorical_cols)} Categorical columns: {categorical_cols}")
print(f"{len(numerical_cols)} Numerical columns: {numerical_cols}")
print(f"{len(binary_cols)} Binary columns: {binary_cols}")

n_feat = len(categorical_cols) + len(numerical_cols) + len(binary_cols)
selected_features = categorical_cols + numerical_cols + binary_cols
print(f"\n{n_feat} Total selected columns: {selected_features}")
```

```
8 Categorical columns: ['job', 'marital', 'education', 'age_group', 'balance_bucket', 'campai
2 Numerical columns: ['day', 'previous']
11 Binary columns: ['pdays_was_contacted', 'default', 'housing', 'loan', 'poutcome_success',

21 Total selected columns: ['job', 'marital', 'education', 'age_group', 'balance_bucket', 'ca
```

Interpretation: The categorical features are unordered and will go through `OneHotEncoder`. The binary features won't be preprocessed in any way since, by construction, they're either 0 or 1. The 2 numerical features will go through `StandardScaler`.

2. Define the preprocessing transformer

Create a preprocessing pipeline that one-hot encodes categorical features and scales numeric/binary features before modeling.

```
preprocessor = ColumnTransformer(
    transformers=[
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_cols),
        ("num", StandardScaler(), numerical_cols),
        ("bin", "passthrough", binary_cols),
    ]
)

print(preprocessor)
```

```
ColumnTransformer(transformers=[('cat', OneHotEncoder(handle_unknown='ignore'),
                                 ['job', 'marital', 'education', 'age_group',
```

```
                                  'balance_bucket', 'campaign_bucket',
                                  'pdays_recency_bucket', 'month']),
                                ('num', StandardScaler(), ['day', 'previous']),
                                ('bin', 'passthrough',
                                 ['pdays_was_contacted', 'default', 'housing',
                                  'loan', 'poutcome_success',
                                  'contact_is_unknown', 'poutcome_is_unknown',
                                  'multiple_contacts_current_campaign',
                                  'previous_and_many_current_contacts',
                                  'many_contacts_and_housing_loan',
                                  'both_housing_and_personal_loan'])])
```

Interpretation: `handle_unknown="ignore"` in `OneHotEncoder` ensures unseen category levels at inference time produce an all-zeros row rather than raising an error. The `ColumnTransformer` is fit only on training data (in Section 3.3) to prevent leaking validation or test distribution information into the preprocessing statistics. By construction, the binary columns won't have any missing values and as such no preprocessing is needed.

3. Split the data and apply preprocessing

Create a initial stratified 80/20 train + validation / test split. Fit the preprocessing pipeline on the training set only and transform the test set.

Afterwards we'll further split the train + validation set into stratified training and validation sets using an 80/20 split. But for the preprocessing and cluster segmentation we'll work with train + validation

```python
y = df_[["target"]].copy()
X = df_[selected_features].copy()

print("\nX shape:", X.shape)
display(X.head(3))

print("\ny shape:", y.shape)
```

```
X shape: (45211, 21)
```

	job	marital	education	age_group	balance_bucket	campaign_bucket	pdays_recency_b
0	management	married	tertiary	51+	high	1	no_previous_con

	job	marital	education	age_group	balance_bucket	campaign_bucket	pdays_recency_b
1	technician	single	secondary	36-50	low	1	no_previous_con
2	entrepreneur	married	secondary	26-35	low	1	no_previous_con

```
y shape: (45211, 1)
```

```
nb = "02"
data_dir = Path("data")

df_path = data_dir / f"{nb}_X_raw.parquet"
X.to_parquet(df_path, index=False)
print(f"Saved X raw: {df_path}")

df_path = data_dir / f"{nb}_y_raw.parquet"
y.to_parquet(df_path, index=False)
print(f"\nSaved y raw: {df_path}")
```

```
Saved X raw: data/02_X_raw.parquet

Saved y raw: data/02_y_raw.parquet
```

```
X_train_val, X_test, y_train_val, y_test = train_test_split(
    X, y, test_size=test_size, random_state=RANDOM_STATE, stratify=y
)

print("\nX train + validation:", X_train_val.shape)
display(X_train_val.head(3))
assert X_train_val.shape[0] == y_train_val.shape[0]

print("\nX test:", X_test.shape)
display(X_test.head(3))
assert X_test.shape[0] == y_test.shape[0]
```

```
X train + validation: (36168, 21)
```

	job	marital	education	age_group	balance_bucket	campaign_bucket	pdays_recency
37311	housemaid	divorced	secondary	36-50	medium	1	medium
28102	technician	single	unknown	36-50	medium	2-3	no_previous_
1792	blue-collar	married	primary	36-50	low	2-3	no_previous_

```
X test: (9043, 21)
```

	job	marital	education	age_group	balance_bucket	campaign_bucket	pdays_recen
28909	management	married	tertiary	36-50	medium	1	no_previous_
45032	blue-collar	single	secondary	26-35	high	1	medium
44779	retired	married	unknown	51+	high	2-3	medium

```python
nb = "02"
data_dir = Path("data")

# Train + Validation
df_path = data_dir / f"{nb}_X_train_val_raw.parquet"
X_train_val.to_parquet(df_path, index=False)
print(f"Saved X_train_val raw: {df_path}")

df_path = data_dir / f"{nb}_y_train_val_raw.parquet"
y_train_val.to_parquet(df_path, index=False)
print(f"Saved y_train_val raw: {df_path}")

# Test
df_path = data_dir / f"{nb}_X_test_raw.parquet"
X_test.to_parquet(df_path, index=False)
print(f"\nSaved X_test raw: {df_path}")

df_path = data_dir / f"{nb}_y_test_raw.parquet"
y_test.to_parquet(df_path, index=False)
print(f"Saved y_test raw: {df_path}")
```

```
Saved X_train_val raw: data/02_X_train_val_raw.parquet
Saved y_train_val raw: data/02_y_train_val_raw.parquet

Saved X_test raw: data/02_X_test_raw.parquet
Saved y_test raw: data/02_y_test_raw.parquet
```

```python
# Fit preprocessing only on train to avoid leakage.
X_train_val_pre = preprocessor.fit_transform(X_train_val)
X_test_pre = preprocessor.transform(X_test)
```

```python
feature_names = preprocessor.get_feature_names_out()

X_train_val_pre = pd.DataFrame(
    X_train_val_pre, columns=feature_names, index=X_train_val.index
)

X_test_pre = pd.DataFrame(X_test_pre, columns=feature_names, index=X_test.index)
```

```python
print(f"Raw train + validation: {X_train_val.shape}")
print(f"Preprocessed train + validation: {X_train_val_pre.shape}")
display(X_train_val_pre.head(3))

print(f"\nRaw test: {X_test.shape}")
print(f"Preprocessed test: {X_test_pre.shape}")
display(X_test_pre.head(3))
```

```
Raw train + validation: (36168, 21)
Preprocessed train + validation: (36168, 59)
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__j
37311	0.0	0.0	0.0	1.0	0.0
28102	0.0	0.0	0.0	0.0	0.0
1792	0.0	1.0	0.0	0.0	0.0

```
Raw test: (9043, 21)
Preprocessed test: (9043, 59)
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__j
28909	0.0	0.0	0.0	0.0	1.0
45032	0.0	1.0	0.0	0.0	0.0
44779	0.0	0.0	0.0	0.0	0.0

Interpretation: In this section we preprocessed the data with the train + validation sets and transformed the test set with it. Since we chose to work with OHE, we can see that the number of features increased from 21 to 59

4. Cluster Analysis

Run a K-Means Clustering algorithm on the following features: `age`, `marital`, `education`, `job`, `balance`, `housing` and `loan`

```
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import (
    silhouette_score,
    davies_bouldin_score,
    calinski_harabasz_score,
)
import re
```

```
feature_names = preprocessor.get_feature_names_out()

keywords = [
    "_job",
    "_marital",
    "_education",
    "_age_group",
    "_balance_bucket",
    "_housing",
    "_loan",
]

exclude_token = ["and", "unknown"]

cluster_columns = [
    col
    for col in feature_names
    if any(kw in col for kw in keywords)
    and not any(token in col for token in exclude_token)
]

cluster_columns
```

```
['cat__job_admin.',
 'cat__job_blue-collar',
 'cat__job_entrepreneur',
 'cat__job_housemaid',
 'cat__job_management',
```

```
 'cat__job_retired',
 'cat__job_self-employed',
 'cat__job_services',
 'cat__job_student',
 'cat__job_technician',
 'cat__job_unemployed',
 'cat__marital_divorced',
 'cat__marital_married',
 'cat__marital_single',
 'cat__education_primary',
 'cat__education_secondary',
 'cat__education_tertiary',
 'cat__age_group_18-25',
 'cat__age_group_26-35',
 'cat__age_group_36-50',
 'cat__age_group_51+',
 'cat__balance_bucket_high',
 'cat__balance_bucket_low',
 'cat__balance_bucket_medium',
 'cat__balance_bucket_negative',
 'bin__housing',
 'bin__loan']
```

@MATTHEW EXPLAIN BRIEFLY WHY WE CHOSE THESE FEATURES (I REMOVED DEFAULT)

```
K_RANGE = range(2, 9)
RANDOM_STATE = RANDOM_STATE

X_train_cluster = X_train_val_pre[list(cluster_columns)].values.astype(float)
X_test_cluster = X_test_pre[list(cluster_columns)].values.astype(float)

print(f"Train + Validation shape: {X_train_cluster.shape}")
print(f"Test shape:   {X_test_cluster.shape}")
```

```
Train + Validation shape: (36168, 27)
Test shape:   (9043, 27)
```

```
pca = PCA(n_components=2, random_state=RANDOM_STATE)
X_train_cluster_pca = pca.fit_transform(X_train_cluster)

print(f"PCA explained variance: {pca.explained_variance_ratio_.sum():.1%}")
```

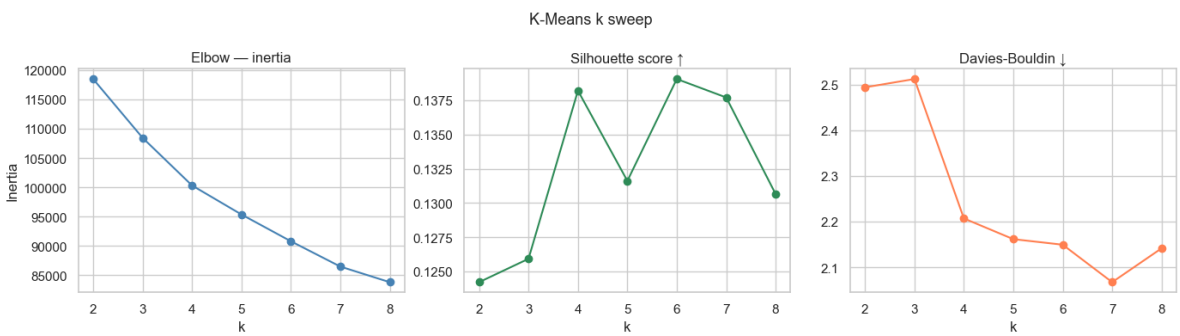
```
PCA explained variance: 27.0%
```

```python
sweep_results = []

for k in K_RANGE:
    km = KMeans(n_clusters=k, n_init=10, random_state=RANDOM_STATE)
    labels = km.fit_predict(X_train_cluster)

    sil = silhouette_score(X_train_cluster, labels)
    db = davies_bouldin_score(X_train_cluster, labels)
    ch = calinski_harabasz_score(X_train_cluster, labels)

    sweep_results.append(
        {
            "k": k,
            "inertia": km.inertia_,
            "silhouette": sil,
            "davies_bouldin": db,
            "calinski_harabasz": ch,
        }
    )
    print(
        f"k={k} | inertia={km.inertia_:,.0f} | silhouette={sil:.4f} | DB={db:.4f} | CH={ch:.
    )

sweep_df = pd.DataFrame(sweep_results)
```

```
k=2 | inertia=118,473 | silhouette=0.1242 | DB=2.4938 | CH=4696.4
k=3 | inertia=108,444 | silhouette=0.1259 | DB=2.5121 | CH=4237.5
k=4 | inertia=100,316 | silhouette=0.1382 | DB=2.2069 | CH=4030.5
k=5 | inertia=95,364 | silhouette=0.1316 | DB=2.1616 | CH=3649.3
k=6 | inertia=90,794 | silhouette=0.1390 | DB=2.1494 | CH=3430.3
k=7 | inertia=86,478 | silhouette=0.1377 | DB=2.0674 | CH=3302.0
k=8 | inertia=83,813 | silhouette=0.1306 | DB=2.1415 | CH=3084.4
```

```python
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

axes[0].plot(sweep_df["k"], sweep_df["inertia"], marker="o", color="steelblue")
axes[0].set_title("Elbow - inertia")
axes[0].set_xlabel("k")
axes[0].set_ylabel("Inertia")
```

```
axes[1].plot(sweep_df["k"], sweep_df["silhouette"], marker="o", color="seagreen")
axes[1].set_title("Silhouette score ↑")
axes[1].set_xlabel("k")

axes[2].plot(sweep_df["k"], sweep_df["davies_bouldin"], marker="o", color="coral")
axes[2].set_title("Davies-Bouldin ↓")
axes[2].set_xlabel("k")

plt.suptitle("K-Means k sweep", fontsize=13)
plt.tight_layout()
plt.show()
```



@MATTHEW ADD INTERPRETATION FOR THESE PLOTS

AND EXPLAIN WHY WE CHOSE K=4 (be creative)

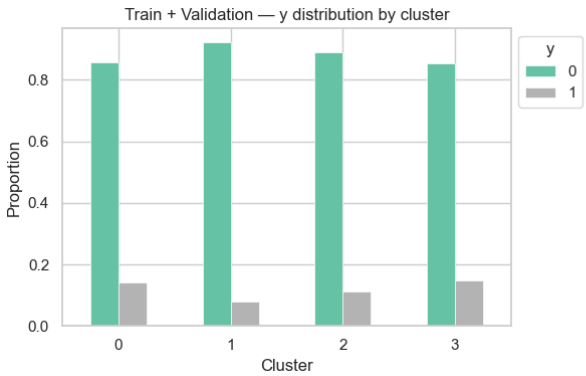
```
chosen_k = 4

km_final = KMeans(n_clusters=chosen_k, n_init=10, random_state=RANDOM_STATE)
train_labels = km_final.fit_predict(X_train_cluster)
test_labels = km_final.predict(X_test_cluster)
```

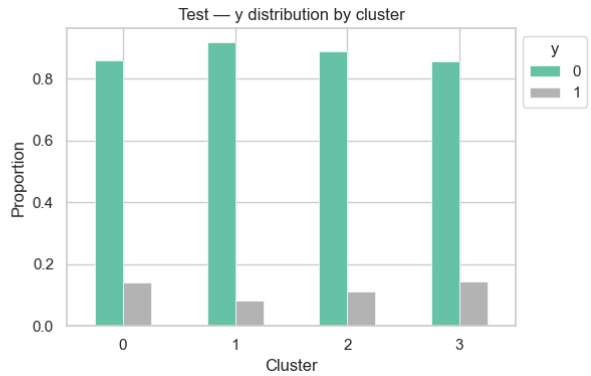
```
metrics = {
    "Silhouette ↑": silhouette_score(X_train_cluster, train_labels),
    "Davies-Bouldin ↓": davies_bouldin_score(X_train_cluster, train_labels),
    "Calinski-Harabasz ↑": calinski_harabasz_score(X_train_cluster, train_labels),
}

print(f"Final model - k = {chosen_k} (train)")
for name, val in metrics.items():
    print(f"  {name:<25s} {val:.4f}")
```

```
Final model - k = 4 (train)
  Silhouette ↑              0.1382
  Davies-Bouldin ↓          2.2069
  Calinski-Harabasz ↑       4030.5211
```

@MATTHEW ADD INTERPRETATION FOR THIS

```python
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

for ax, labels, y, title in zip(
    axes,
    [train_labels, test_labels],
    [y_train_val, y_test],
    ["Train + Validation", "Test"],
):
    plot_df = pd.DataFrame(
        {"cluster": np.array(labels).ravel(), "y": np.array(y).ravel()}
    )

    cluster_props = (
        plot_df.groupby("cluster")["y"]
        .value_counts(normalize=True)
        .rename("proportion")
        .reset_index()
    )

    pivot = cluster_props.pivot(
        index="cluster", columns="y", values="proportion"
    ).fillna(0)

    pivot.plot(kind="bar", ax=ax, colormap="Set2", edgecolor="white", linewidth=0.5)
    ax.set_title(f"{title} - y distribution by cluster")
    ax.set_xlabel("Cluster")
    ax.set_ylabel("Proportion")
    ax.set_xticklabels(ax.get_xticklabels(), rotation=0)
    ax.legend(title="y", bbox_to_anchor=(1, 1))

plt.tight_layout()
plt.show()
```





@MATTHEW DESCRIBE WHAT WE SEE

```
cluster_feat = "cluster"

X_train_val_clustered = X_train_val_pre.copy()
X_train_val_clustered[cluster_feat] = train_labels
print(f"Original preprocessed train + validation: {X_train_val_pre.shape}")
print(f"Clustered preprocessed train + validation: {X_train_val_clustered.shape}")

X_test_clustered = X_test_pre.copy()
X_test_clustered[cluster_feat] = test_labels
print(f"\nOriginal preprocessed test: {X_test_pre.shape}")
print(f"Clustered preprocessed test: {X_test_clustered.shape}")
```

```
Original preprocessed train + validation: (36168, 59)
Clustered preprocessed train + validation: (36168, 60)

Original preprocessed test: (9043, 59)
Clustered preprocessed test: (9043, 60)
```

```
print(
    f"Train + Validation cluster counts:\n"
    f"{X_train_val_clustered[cluster_feat].value_counts(normalize=True).sort_index()}"
)

print(
    f"\nTest cluster counts:\n"
    f"{X_test_clustered[cluster_feat].value_counts(normalize=True).sort_index()}"
)
```

```
Train + Validation cluster counts:
cluster
0    0.174574
1    0.307261
2    0.242701
3    0.275464
Name: proportion, dtype: float64

Test cluster counts:
cluster
0    0.178259
1    0.306204
2    0.243393
3    0.272144
Name: proportion, dtype: float64
```

@MATTHEW MENTION HOW WE HAVE BALANCE IN BOTH DATASETS

We now save the preprocessed datasets

```
print(f"X train + validation preprocessed: {X_train_val_clustered.shape}")
print(f"y train + validation preprocessed: {y_train_val.shape}")
display(X_train_val_clustered.head(2))

print(f"\nX test preprocessed: {X_test_clustered.shape}")
print(f"y test preprocessed: {y_test.shape}")
display(X_test_clustered.head(2))
```

```
X train + validation preprocessed: (36168, 60)
y train + validation preprocessed: (36168, 1)
```

	cat___job_admin.	cat___job_blue-collar	cat___job_entrepreneur	cat___job_housemaid	cat___j
37311	0.0	0.0	0.0	1.0	0.0
28102	0.0	0.0	0.0	0.0	0.0

```
X test preprocessed: (9043, 60)
y test preprocessed: (9043, 1)
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__j
28909	0.0	0.0	0.0	0.0	1.0
45032	0.0	1.0	0.0	0.0	0.0

```
nb = "02"
data_dir = Path("data")

# Train + Validation
df_path = data_dir / f"{nb}_X_train_val.parquet"
X_train_val_clustered.to_parquet(df_path, index=False)
print(f"Saved X_train_val preprocessed + clustered: {df_path}")

df_path = data_dir / f"{nb}_y_train_val.parquet"
y_train_val.to_parquet(df_path, index=False)
print(f"Saved y_train_val preprocessed + clustered: {df_path}")

# Test
df_path = data_dir / f"{nb}_X_test.parquet"
X_test_clustered.to_parquet(df_path, index=False)
print(f"\nSaved X_test preprocessed + clustered: {df_path}")

df_path = data_dir / f"{nb}_y_test.parquet"
y_test.to_parquet(df_path, index=False)
print(f"Saved y_test preprocessed + clustered: {df_path}")
```

```
Saved X_train_val: data/02_X_train_val.parquet
Saved y_train_val raw: data/02_y_train_val.parquet

Saved X_test raw: data/02_X_test.parquet
Saved y_test raw: data/02_y_test.parquet
```

5. Create datasets per cluster

Given that we now have our preprocessed and clustered data, we'll now create our train, validation and test sets for: 1. Entire dataset 2. Each cluster

These will be the datasets that we'll use during training in the following notebook

```python
nb = "02"
data_dir = Path("data")

# Train + Validation
df_path = data_dir / f"{nb}_X_train_val.parquet"
X_train_val_ = pd.read_parquet(df_path)
print(f"X train + validation: {X_train_val_.shape}")

df_path = data_dir / f"{nb}_y_train_val.parquet"
y_train_val_ = pd.read_parquet(df_path)
print(f"y train + validation: {y_train_val_.shape}")

# Test
df_path = data_dir / f"{nb}_X_test.parquet"
X_test_ = pd.read_parquet(df_path)
print(f"\nX test: {X_test_.shape}")

df_path = data_dir / f"{nb}_y_test.parquet"
y_test_ = pd.read_parquet(df_path)
print(f"y test: {y_test_.shape}")
```

```
X train + validation: (36168, 60)
y train + validation: (36168, 1)

X test: (9043, 60)
y test: (9043, 1)
```

```python
X_train, X_validation, y_train, y_validation = train_test_split(
    X_train_val_,
    y_train_val_,
    test_size=test_size,
    random_state=RANDOM_STATE,
    stratify=y_train_val_,
)

print("\nX train:", X_train.shape)
assert X_train.shape[0] == y_train.shape[0]

print("X validation:", X_validation.shape)
assert X_validation.shape[0] == y_validation.shape[0]
```

```
assert X_train.shape[0] + X_validation.shape[0] == X_train_val_.shape[0]
assert y_train.shape[0] + y_validation.shape[0] == y_train_val_.shape[0]
```

```
X train: (28934, 60)
X validation: (7234, 60)
```

```
nb = "02"
data_dir = Path("data")

# Train
df_path = data_dir / f"{nb}_X_train.parquet"
X_train.to_parquet(df_path, index=False)
print(f"Saved X_train: {df_path}")

df_path = data_dir / f"{nb}_y_train.parquet"
y_train.to_parquet(df_path, index=False)
print(f"Saved y_train: {df_path}")

# Validation
df_path = data_dir / f"{nb}_X_validation.parquet"
X_validation.to_parquet(df_path, index=False)
print(f"Saved X_validation: {df_path}")

df_path = data_dir / f"{nb}_y_validation.parquet"
y_validation.to_parquet(df_path, index=False)
print(f"Saved y_validation: {df_path}")
```

```
Saved X_train: data/02_X_train.parquet
Saved y_train: data/02_y_train.parquet
Saved X_validation: data/02_X_validation.parquet
Saved y_validation: data/02_y_validation.parquet
```

We'll use these datasets for the global model (the one that includes all the segments)

Datasets per segment

```python
nb = "02"
data_dir = Path("data")

# Load datasets
X_train = pd.read_parquet(data_dir / f"{nb}_X_train.parquet")
y_train = pd.read_parquet(data_dir / f"{nb}_y_train.parquet")

X_val = pd.read_parquet(data_dir / f"{nb}_X_validation.parquet")
y_val = pd.read_parquet(data_dir / f"{nb}_y_validation.parquet")

X_test = pd.read_parquet(data_dir / f"{nb}_X_test.parquet")
y_test = pd.read_parquet(data_dir / f"{nb}_y_test.parquet")
```

```python
def split_and_save_by_cluster(X, y, split_name):
    print(f"{split_name}")
    for cluster_id in sorted(X["cluster"].unique()):
        mask = X["cluster"] == cluster_id

        X_subset = X.loc[mask].copy()
        y_subset = y.loc[mask].copy()

        X_subset.to_parquet(data_dir / f"{nb}_X_{split_name}_c{cluster_id}.parquet")
        y_subset.to_parquet(data_dir / f"{nb}_y_{split_name}_c{cluster_id}.parquet")

        print(
            f"\tCluster {cluster_id}: {X_subset.shape} ({X_subset.shape[0] / X.shape[0] :.2%
        )
```

```python
split_and_save_by_cluster(X_train, y_train, "train")
print("\n")
split_and_save_by_cluster(X_val, y_val, "validation")
print("\n")
split_and_save_by_cluster(X_test, y_test, "test")
```

```
train
    Cluster 0: (5043, 60) (17.43%)
    Cluster 1: (8847, 60) (30.58%)
    Cluster 2: (7058, 60) (24.39%)
    Cluster 3: (7986, 60) (27.60%)
```

```
validation
    Cluster 0: (1271, 60) (17.57%)
    Cluster 1: (2266, 60) (31.32%)
    Cluster 2: (1720, 60) (23.78%)
    Cluster 3: (1977, 60) (27.33%)


test
    Cluster 0: (1612, 60) (17.83%)
    Cluster 1: (2769, 60) (30.62%)
    Cluster 2: (2201, 60) (24.34%)
    Cluster 3: (2461, 60) (27.21%)
```

3.4. Inspect the transformed feature names

Retrieve the final feature names after one-hot encoding and scaling to see the exact modeling input.

```python
feature_names = preprocessor.get_feature_names_out()

print("Number of transformed features:", len(feature_names))
print("\nFirst 30 feature names:")
print(feature_names[:30])
```

```
Number of transformed features: 59

First 30 feature names:
['cat__job_admin.' 'cat__job_blue-collar' 'cat__job_entrepreneur'
 'cat__job_housemaid' 'cat__job_management' 'cat__job_retired'
 'cat__job_self-employed' 'cat__job_services' 'cat__job_student'
 'cat__job_technician' 'cat__job_unemployed' 'cat__job_unknown'
 'cat__marital_divorced' 'cat__marital_married' 'cat__marital_single'
 'cat__education_primary' 'cat__education_secondary'
 'cat__education_tertiary' 'cat__education_unknown' 'cat__age_group_18-25'
 'cat__age_group_26-35' 'cat__age_group_36-50' 'cat__age_group_51+'
 'cat__balance_bucket_high' 'cat__balance_bucket_low'
 'cat__balance_bucket_medium' 'cat__balance_bucket_negative'
 'cat__campaign_bucket_1' 'cat__campaign_bucket_2-3'
 'cat__campaign_bucket_4+']
```

Interpretation: OHE expands the 7 categorical columns into many binary indicator columns (one per distinct category level), while the 14 numeric/binary columns pass through with a

num__ prefix. The total of 56 processed features (vs. 21 raw inputs) is the final dimensionality that downstream models will consume.

3.5. Convert the transformed arrays to data frames

Convert the preprocessed training, validation, and test sets into DataFrames with feature names for easier inspection.

```
# Convert sparse/dense transformed matrices into labeled DataFrames for inspection.
X_train_processed_df = pd.DataFrame(
    (
        X_train_processed.toarray()
        if hasattr(X_train_processed, "toarray")
        else X_train_processed
    ),
    columns=feature_names,
    index=X_train.index,
)

X_val_processed_df = pd.DataFrame(
    (
        X_val_processed.toarray()
        if hasattr(X_val_processed, "toarray")
        else X_val_processed
    ),
    columns=feature_names,
    index=X_val.index,
)

X_test_processed_df = pd.DataFrame(
    (
        X_test_processed.toarray()
        if hasattr(X_test_processed, "toarray")
        else X_test_processed
    ),
    columns=feature_names,
    index=X_test.index,
)

print("X_train_processed_df shape:", X_train_processed_df.shape)
print("X_val_processed_df shape:", X_val_processed_df.shape)
print("X_test_processed_df shape:", X_test_processed_df.shape)
```

```
display(X_train_processed_df.head())
```

```
X_train_processed_df shape: (28934, 59)
X_val_processed_df shape: (7234, 59)
X_test_processed_df shape: (9043, 59)
```

	cat__job_admin.	cat__job_blue-collar	cat__job_entrepreneur	cat__job_housemaid	cat__j
30140	0.0	0.0	0.0	0.0	0.0
43748	1.0	0.0	0.0	0.0	0.0
22363	0.0	0.0	0.0	0.0	0.0
4603	0.0	1.0	0.0	0.0	0.0
20619	0.0	0.0	0.0	0.0	0.0

3.6. Check the transformed data

Run a quick sanity check on the preprocessed training, validation, and test sets to confirm there are no missing or constant features.

```
# Missing-value checks ensure preprocessing produced complete model inputs.
print(
    "Missing values in X_train_processed_df:", X_train_processed_df.isna().sum().sum()
)
print("Missing values in X_val_processed_df:", X_val_processed_df.isna().sum().sum())
print("Missing values in X_test_processed_df:", X_test_processed_df.isna().sum().sum())

# Constant columns may carry little signal for some model families.
constant_cols = [
    c for c in X_train_processed_df.columns if X_train_processed_df[c].nunique() == 1
]

print("Number of constant columns:", len(constant_cols))
print("Constant columns:")
print(constant_cols)
```

```
Missing values in X_train_processed_df: 0
Missing values in X_val_processed_df: 0
Missing values in X_test_processed_df: 0
Number of constant columns: 0
```

```
Constant columns:
[]
```

Interpretation: Zero missing values across train, validation, and test means the preprocessing pipeline is producing complete features. If the number of constant columns is small, most transformed features retain variability and are likely informative for modeling.

3.7 Cluster Analysis

Run a K-Means Clustering algorithm on the following features: age, marital, education, job, balance, housing, loan, and default

```
cluster_columns = (
    "cat__age_group_18-25",
    "cat__age_group_26-35",
    "cat__age_group_36-50",
    "cat__age_group_51+",
    "cat__marital_divorced",
    "cat__marital_married",
    "cat__marital_single",
    "cat__education_primary",
    "cat__education_secondary",
    "cat__education_tertiary",
    "cat__job_admin.",
    "cat__job_blue-collar",
    "cat__job_entrepreneur",
    "cat__job_housemaid",
    "cat__job_management",
    "cat__job_retired",
    "cat__job_self-employed",
    "cat__job_services",
    "cat__job_student",
    "cat__job_technician",
    "cat__job_unemployed",
    "cat__balance_bucket_high",
    "cat__balance_bucket_low",
    "cat__balance_bucket_medium",
    "cat__balance_bucket_negative",
    "num__housing_binary",
    "num__loan_binary",
    "num__default_binary",
)
```

```python
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import (
    silhouette_score,
    davies_bouldin_score,
    calinski_harabasz_score,
)
```

```python
K_RANGE = range(2, 9)
RANDOM_STATE = 42

X_train_val_processed_df = pd.concat([X_train_processed_df, X_val_processed_df], axis=0)
y_train_val = pd.concat([y_train, y_val], axis=0)

X_train_cluster = X_train_val_processed_df[list(cluster_columns)].values.astype(float)
X_test_cluster = X_test_processed_df[list(cluster_columns)].values.astype(float)

print(f"Train shape: {X_train_cluster.shape}")
print(f"Val shape:   {X_test_cluster.shape}")
```

```
Train shape: (36168, 28)
Val shape:   (9043, 28)
```

```python
pca = PCA(n_components=2, random_state=RANDOM_STATE)
X_train_cluster_pca = pca.fit_transform(X_train_cluster)

print(f"PCA explained variance: {pca.explained_variance_ratio_.sum():.1%}")
```

```
PCA explained variance: 33.7%
```

```
/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/decomposition/_pca.py:6
  C = X.T @ X
/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/decomposition/_pca.py:6
  C = X.T @ X
/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/decomposition/_pca.py:6
  C = X.T @ X
/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/decomposition/_base.py
  X_transformed = X @ self.components_.T
/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/decomposition/_base.py
  X_transformed = X @ self.components_.T
```

```
/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/decomposition/_base.py
  X_transformed = X @ self.components_.T
```

```
sweep_results = []

for k in K_RANGE:
    km = KMeans(n_clusters=k, n_init=10, random_state=RANDOM_STATE)
    labels = km.fit_predict(X_train_cluster)

    sil = silhouette_score(X_train_cluster, labels)
    db = davies_bouldin_score(X_train_cluster, labels)
    ch = calinski_harabasz_score(X_train_cluster, labels)

    sweep_results.append(
        {
            "k": k,
            "inertia": km.inertia_,
            "silhouette": sil,
            "davies_bouldin": db,
            "calinski_harabasz": ch,
        }
    )
    print(
        f"k={k} | inertia={km.inertia_:,.0f} | silhouette={sil:.4f} | DB={db:.4f} | CH={ch:.
    )

sweep_df = pd.DataFrame(sweep_results)
```

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/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/utils/extmath.py:203: R
  ret = a @ b
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  ret = a @ b

k=2 | inertia=190,459 | silhouette=0.1859 | DB=2.0894 | CH=7040.9

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```

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  current_pot = closest_dist_sq @ sample_weight

k=3 | inertia=155,250 | silhouette=0.2025 | DB=1.5184 | CH=8419.6

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  current_pot = closest_dist_sq @ sample_weight

k=4 | inertia=125,968 | silhouette=0.2349 | DB=1.3718 | CH=9719.8

/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/utils/extmath.py:203: R
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/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/cluster/_kmeans.py:237
  current_pot = closest_dist_sq @ sample_weight

k=5 | inertia=119,226 | silhouette=0.1783 | DB=1.7357 | CH=8213.1

/Users/matthewfischer/miniforge3/lib/python3.12/site-packages/sklearn/utils/extmath.py:203: R
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k=6 | inertia=112,961 | silhouette=0.1595 | DB=1.9255 | CH=7335.9

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k=7 | inertia=107,784 | silhouette=0.1647 | DB=1.9610 | CH=6696.1

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k=8 | inertia=102,982 | silhouette=0.1622 | DB=1.9716 | CH=6247.9

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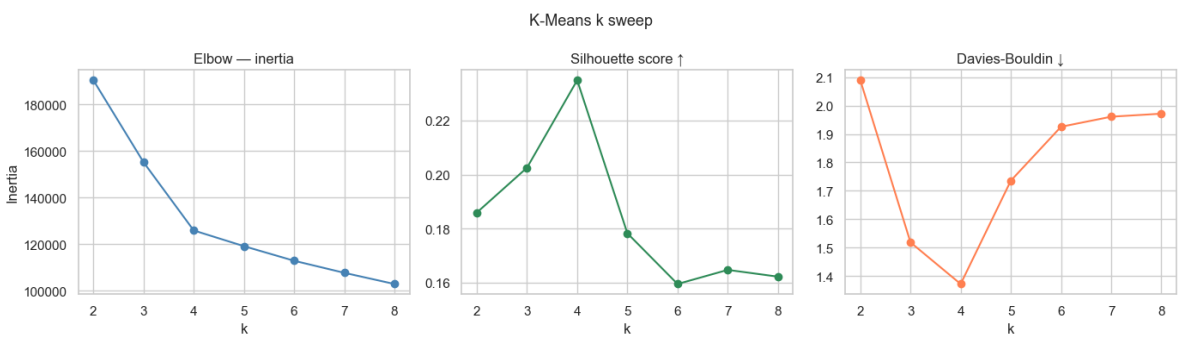
```
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

axes[0].plot(sweep_df["k"], sweep_df["inertia"], marker="o", color="steelblue")
axes[0].set_title("Elbow - inertia")
axes[0].set_xlabel("k")
axes[0].set_ylabel("Inertia")

axes[1].plot(sweep_df["k"], sweep_df["silhouette"], marker="o", color="seagreen")
axes[1].set_title("Silhouette score ↑")
axes[1].set_xlabel("k")

axes[2].plot(sweep_df["k"], sweep_df["davies_bouldin"], marker="o", color="coral")
axes[2].set_title("Davies-Bouldin ↓")
axes[2].set_xlabel("k")

plt.suptitle("K-Means k sweep", fontsize=13)
plt.tight_layout()
plt.show()
```




```
chosen_k = int(sweep_df.loc[sweep_df["silhouette"].idxmax(), "k"])

print(f"Chosen k: {chosen_k}")
```

```
Chosen k: 4
```

```
km_final = KMeans(n_clusters=chosen_k, n_init=10, random_state=RANDOM_STATE)
train_labels = km_final.fit_predict(X_train_cluster)
test_labels = km_final.predict(X_test_cluster)

print(f"Train cluster counts:\n{pd.Series(train_labels).value_counts().sort_index()}")
print(f"\nTest cluster counts:\n{pd.Series(test_labels).value_counts().sort_index()}")
```

```
Train cluster counts:
0     5568
1    16413
2      647
3    13540
Name: count, dtype: int64

Test cluster counts:
0    1375
1    4048
2     168
3    3452
Name: count, dtype: int64

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```

```python
metrics = {
    "Silhouette ↑": silhouette_score(X_train_cluster, train_labels),
    "Davies-Bouldin ↓": davies_bouldin_score(X_train_cluster, train_labels),
    "Calinski-Harabasz ↑": calinski_harabasz_score(X_train_cluster, train_labels),
}

print(f"Final model - k={chosen_k}")
for name, val in metrics.items():
    print(f"  {name:<25s} {val:.4f}")
```

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Final model - k=4
  Silhouette ↑              0.2349
  Davies-Bouldin ↓          1.3718
  Calinski-Harabasz ↑       9719.7644

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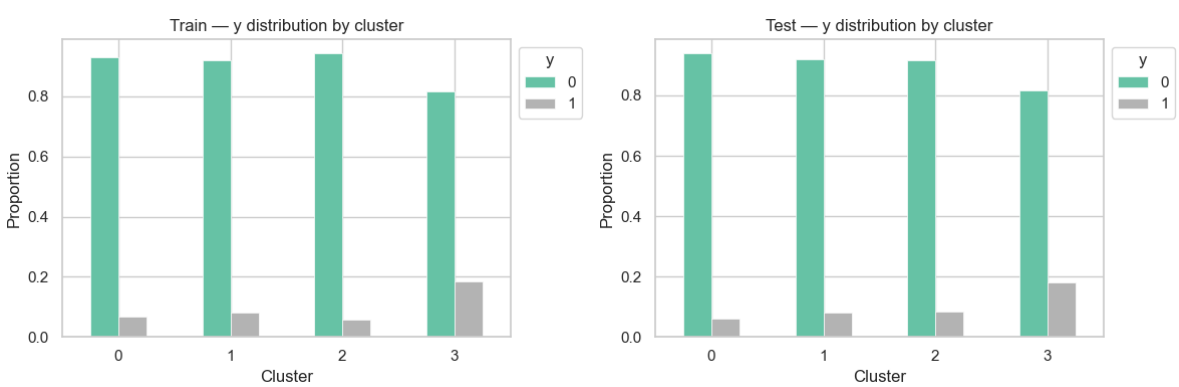
```

```python
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

for ax, labels, y, title in zip(
    axes,
    [train_labels, test_labels],
    [y_train_val, y_test],
    ["Train", "Test"],
):
    plot_df = pd.DataFrame({"cluster": labels, "y": y})
    cluster_props = (
        plot_df.groupby("cluster")["y"]
        .value_counts(normalize=True)
        .rename("proportion")
        .reset_index()
    )
    pivot = cluster_props.pivot(
        index="cluster", columns="y", values="proportion"
    ).fillna(0)
    pivot.plot(kind="bar", ax=ax, colormap="Set2", edgecolor="white", linewidth=0.5)
    ax.set_title(f"{title} - y distribution by cluster")
    ax.set_xlabel("Cluster")
    ax.set_ylabel("Proportion")
    ax.set_xticklabels(ax.get_xticklabels(), rotation=0)
    ax.legend(title="y", bbox_to_anchor=(1, 1))

plt.tight_layout()
plt.show()
```




```python
train_cluster_ohe = pd.get_dummies(train_labels, prefix="cluster").astype(int)
test_cluster_ohe = pd.get_dummies(test_labels, prefix="cluster").astype(int)
```

```python
train_cluster_ohe, test_cluster_ohe = train_cluster_ohe.align(
    test_cluster_ohe, join="left", axis=1, fill_value=0
)

X_train_val_processed_df = X_train_val_processed_df.reset_index(drop=True)
X_test_processed_df = X_test_processed_df.reset_index(drop=True)
train_cluster_ohe = train_cluster_ohe.reset_index(drop=True)
test_cluster_ohe = test_cluster_ohe.reset_index(drop=True)

X_train_val_final = pd.concat([X_train_val_processed_df, train_cluster_ohe], axis=1)
X_test_final = pd.concat([X_test_processed_df, test_cluster_ohe], axis=1)

print(f"Train shape: {X_train_val_final.shape}")
print(f"Test shape: {X_test_final.shape}")
print(f"New columns: {list(train_cluster_ohe.columns)}")

nb = "02"
data_dir = Path("data")
data_dir.mkdir(parents=True, exist_ok=True)

df_path_train = data_dir / f"{nb}_train.parquet"
df_path_test = data_dir / f"{nb}_test.parquet"

X_train_val_final.to_parquet(df_path_train, index=False)
X_test_final.to_parquet(df_path_test, index=False)
```

```
Train shape: (36168, 63)
Test shape: (9043, 63)
New columns: ['cluster_0', 'cluster_1', 'cluster_2', 'cluster_3']
```